

MASLOV ASYMPTOTIC GREEN'S FUNCTIONS IN HETEROGENEOUS ANISOTROPIC ELASTIC MEDIA

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For wave propagation in media that are non-trivially heterogeneous and elastic, the formation of caustics is quite common. Due to the heterogeneity, multi-pathing in the set of characteristics or geometrical rays might occur, while due to anisotropic elasticity the quasi-shear-wave fronts might 'fold'. Both phenomena truly coexist in geophysical applications, and may be hard to separate. It is a well known fact that geometrical ray-theory (GRA) breaks down in such situations since it only considers rays that satisfy Fermat's principle of stationary traveltime. This can introduce singularities in the system of dynamic raytracing equations. The Maslov asymptotics approach can to great extent overcome this problem by taking into account contributions from neighboring rays in an integral over the local wavefront. This will smooth the contributions from sharp features on the wavefront such as caustics and produce an asymptotically correct amplitude. The Maslov amplitude can be found by formulating the system of raytracing equations on the Lagrangian manifold (Maslov & Fedoriuk, 1981). Here, Liouville's theorem ensures the existence of a non-singular volume element that is used to calculate the geometrical spreading. By selecting suitable projections of this volume element from phase-space into physical space, we can construct nonsingular ray Jacobians along raypaths even inside caustics (Chapman & Drummond, 1982).

The raytracing is implemented in 3D for smooth, heterogeneous, anisotropic, elastic media based on the work of Cerveny (1995). The raytracing equations are solved on a Cartesian grid using an adaptive step-length Runge-Kutta scheme with traveltime as the monotonically increasing parameter along the ray. This enables us to take long ray-steps in relatively simple regions of the medium, but carefully step through the more complex parts, maintaining a high order of accuracy. The speed-up factor by this approach is highly variable, but general consensus is that it is significant for most practical applications (Press et al., 1994). In the ray code, the rays are sampled uniformly in phase direction at the source. The integral over slowness in the Maslov representation of the Green's function is evaluated in the frequency domain by binning the slowness vectors at the recording surface and then carry out the summation over slowness with an appropriate smooth amplitude and phase function for each bin. To determine the correct projection of the ray Jacobian to be used in the evaluation of the amplitudes, we find the largest possible Jacobian that can be constructed at any given point along the ray. The final time-domain Green's function is produced by an inverse Fourier transform from the (x, ω) -domain to the (x, t) -domain.

Raytracing combined with Maslov asymptotics provide a flexible tool for computing 3D Green's functions in complex geological regions and heterogeneous anisotropic media. For seismic imaging algorithms to perform satisfactory in such media, where multi-pathing and caustics are common, the computed Green's functions must take these effects into account (de Hoop & Brandsberg-Dahl, 2000). Our implementation, with an adaptive dimension of the Maslov integral, will cut the computational time in regions where the wavefields are simple, but still prove correct high frequency amplitudes in complex areas.

References

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